



**Stan's
Technologies**

Benchmark Report

BoolMinGeo vs PyEDA Boolean Minimization

Experiment Date: 2026-01-13

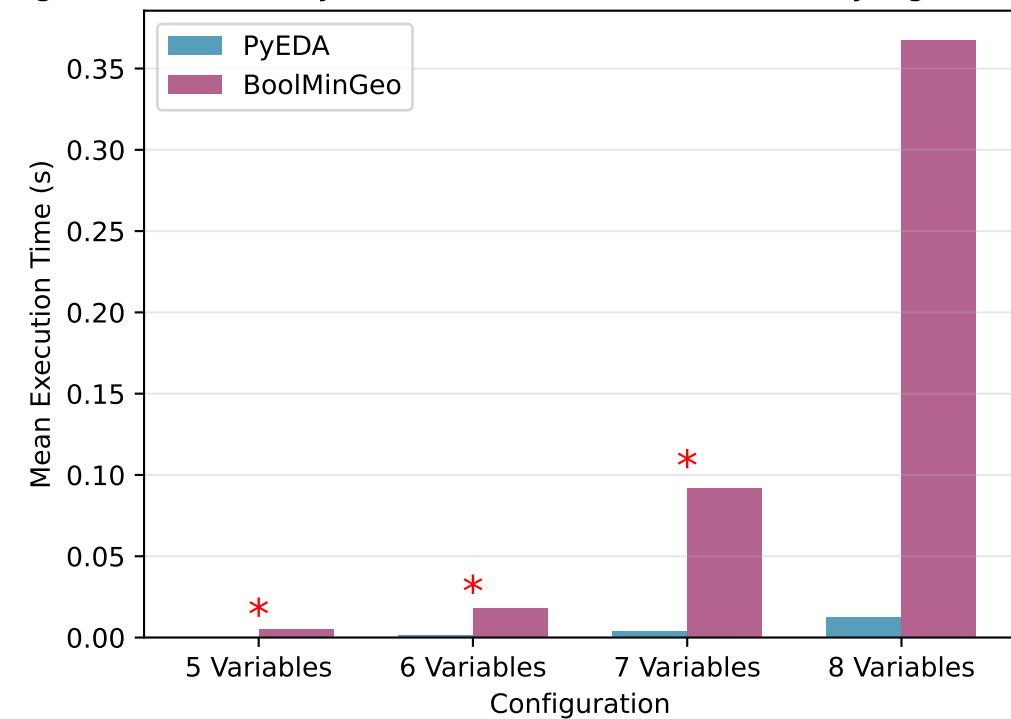
Random Seed: 42

Total Test Cases: 400

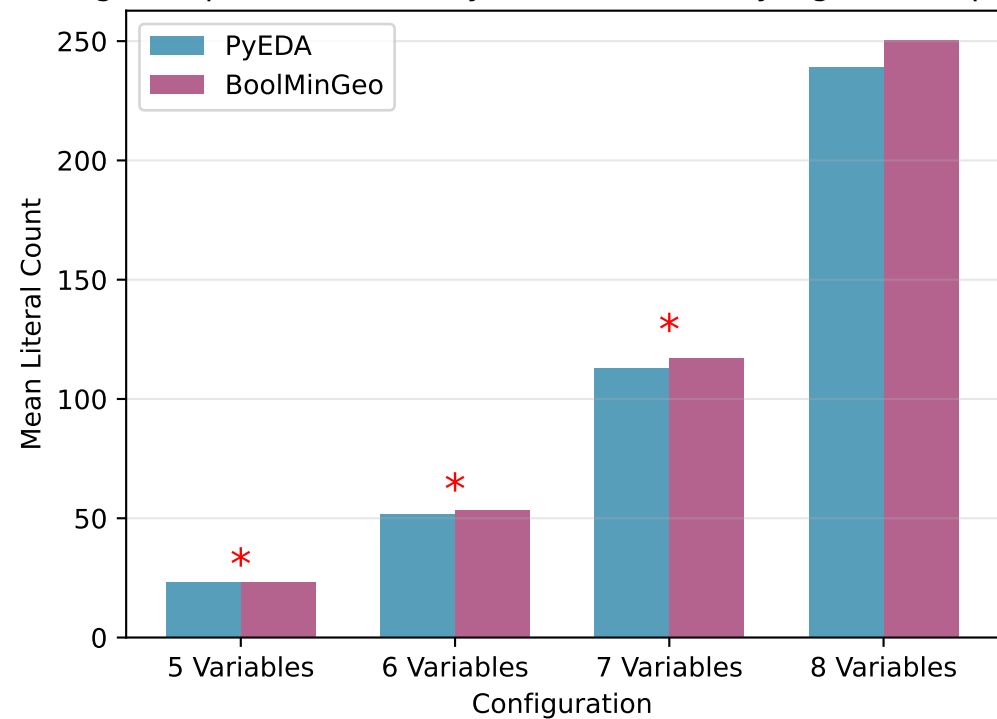
Statistical Significance Level: $\alpha = 0.05$

A Statistical Comparison of Boolean Minimization Algorithms

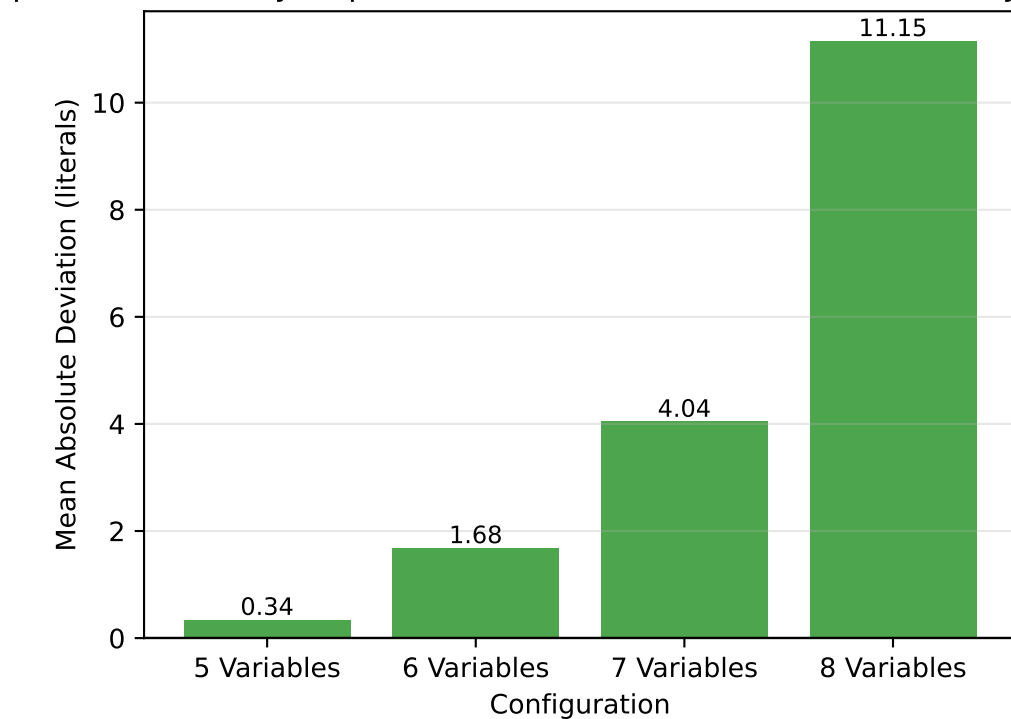
verage Performance by Variable Count\n(* = statistically significant, p < 0.05)



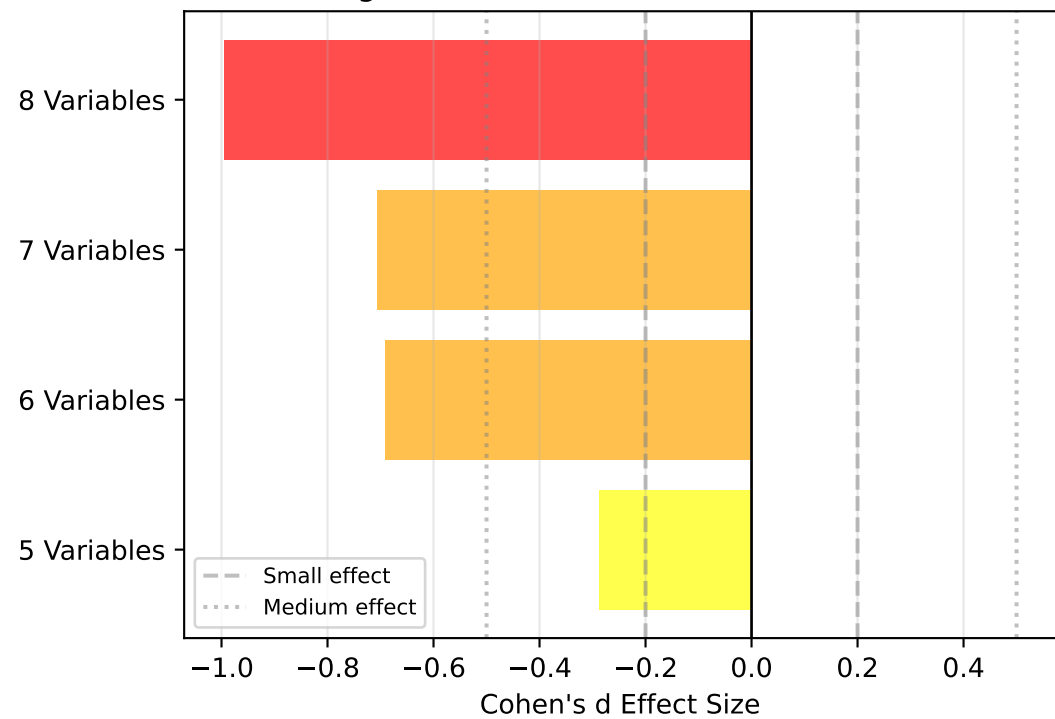
verage Simplification Quality\n(* = statistically significant, p < 0.05)



verage Simplification Quality Gap\n(Green = BoolMinGeo better, Red = PyEDA better)



Effect Size: Simplification Quality
(Negative = BoolMinGeo more minimal)



STATISTICAL ANALYSIS SUMMARY

EXPERIMENT OVERVIEW

=====
Total Test Cases: 400
Configurations Tested: 4 (5-8 variables)
Tests per Config: 10 (sparse, balanced, dense distributions)
Random Seed: 42
Significance Level: $\alpha = 0.05$

5-VARIABLE FUNCTIONS (100 tests, 100 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.000447 s
Mean BoolMinGeo: 0.004940 s
Mean Difference: -0.004493 s
95% CI: [-0.004824, -0.004161]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -2.6891 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 23.08 literals
Mean BoolMinGeo: 23.42 literals
Mean Difference: -0.34 literals
Mean Deviation: 0.34 literals
Quality Gap: 1.47% relative to PyEDA
95% CI: [-0.57, -0.11]
p-value: 0.004944
Statistically Sig: YES
Effect Size (d): -0.2875 (small)
Winner: PyEDA

6-VARIABLE FUNCTIONS (100 tests, 100 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.001358 s
Mean BoolMinGeo: 0.018371 s
Mean Difference: -0.017014 s
95% CI: [-0.018417, -0.015611]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -2.4062 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 51.69 literals
Mean BoolMinGeo: 53.37 literals
Mean Difference: -1.68 literals
Mean Deviation: 1.68 literals
Quality Gap: 3.25% relative to PyEDA
95% CI: [-2.16, -1.20]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -0.6907 (medium)
Winner: PyEDA

7-VARIABLE FUNCTIONS (100 tests, 100 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.004246 s
Mean BoolMinGeo: 0.091984 s
Mean Difference: -0.087737 s
95% CI: [-0.092965, -0.082510]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -3.3300 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 113.01 literals
Mean BoolMinGeo: 117.05 literals
Mean Difference: -4.04 literals
Mean Deviation: 4.04 literals
Quality Gap: 3.57% relative to PyEDA
95% CI: [-5.17, -2.91]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -0.7064 (medium)
Winner: PyEDA

8-VARIABLE FUNCTIONS (100 tests, 100 non-constant)

EXECUTION TIME:
Mean PyEDA: 0.012307 s
Mean BoolMinGeo: 0.367223 s
Mean Difference: -0.354915 s
95% CI: [-0.374261, -0.335570]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -3.6403 (large)
Winner: PyEDA

SIMPLIFICATION QUALITY (Literal Count):
Mean PyEDA: 239.11 literals
Mean BoolMinGeo: 250.26 literals
Mean Difference: -11.15 literals
Mean Deviation: 11.15 literals
Quality Gap: 4.66% relative to PyEDA
95% CI: [-13.37, -8.93]
p-value: 0.000000
Statistically Sig: YES
Effect Size (d): -0.9957 (large)
Winner: PyEDA

CONCLUSIONS

KEY FINDINGS

- 1. PERFORMANCE:
 - Average PyEDA time: 0.004590 s
 - Average BoolMinGeo time: 0.120629 s
 - Difference: -0.116040 s
 - Statistical significance: YES (p = 0.000000)
- 2. SIMPLIFICATION QUALITY:
 - Average PyEDA literals: 106.72
 - Average BoolMinGeo literals: 111.03
 - Mean deviation: 4.30 literals
 - Quality gap: 4.03% relative to PyEDA
 - Statistical significance: YES (p = 0.000000)
- 3. EQUIVALENCE:
 - All 400/400 tests passed logical equivalence checks
 - Both algorithms produce functionally correct results
- 4. EFFECT SIZES:
 - Performance: -0.7715 (medium)
 - Simplification: -0.5627 (medium)

INTERPRETATION

Per-Variable Analysis:

- 5 variables: 1.47% quality gap, PyEDA produces more minimal results (Mean deviation: 0.34 literals)
- 6 variables: 3.25% quality gap, PyEDA produces more minimal results (Mean deviation: 1.68 literals)
- 7 variables: 3.57% quality gap, PyEDA produces more minimal results (Mean deviation: 4.04 literals)
- 8 variables: 4.66% quality gap, PyEDA produces more minimal results (Mean deviation: 11.15 literals)

REPRODUCIBILITY

- Random seed: 42
- All test cases can be reproduced with this seed
- Statistical tests use standard methods (paired t-test, Cohen's d)

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